

Usama Ikram

MS Candidate in Artificial Intelligence · Daegu Gyeongbuk Institute of Science and Technology (DGIST), Republic of Korea

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RESEARCH INTERESTS

Computer vision and multimodal AI, with a focus on adapting large pretrained models to new domains and checking whether they can be trusted where they are used. My MS research covers LoRA adaptation of vision and vision-language models (SAM-family backbones and CLIP), a comparison of a generative and a contrastive vision-language model in accuracy, representation and latency, and learned segmentation and phase reconstruction for quantitative cell analysis in digital holographic microscopy (DHM). Before my MS I built and deployed vision, time-series and LLM-based systems in industry, including on embedded hardware. In a PhD I want to work on vision-language models and large language models for healthcare and biomedical imaging, with emphasis on explainable and trustworthy AI.

EDUCATION

Daegu Gyeongbuk Institute of Science and Technology (DGIST) · Daegu, Republic of Korea Sep 2025 – present
MS in Artificial Intelligence, Department of Artificial Intelligence. GPA 3.86 / 4.30. Expected completion June 2027.

Graduate coursework: Optical Signal Processing (A+), Biomedical Programming (A+), Convex Optimization (A0), Deep Learning Lab (A0), Random Variables and Random Processes, Linear Systems.

National University of Sciences and Technology (NUST) · Rawalpindi, Pakistan Sep 2019 – Jun 2023
Bachelor of Electrical Engineering, College of Electrical and Mechanical Engineering. CGPA 3.27 / 4.00.

Relevant coursework: Machine Learning (A), Computer Vision, Digital Image Processing, Numerical Methods (A), Signals and Systems, Digital Signal Processing, Linear Control Systems, Probability and Statistics, Linear Algebra and ODEs. Final-year project (A): early detection of Alzheimer's disease.

MANUSCRIPTS UNDER REVIEW

All four manuscripts are first-authored and currently under peer review. None has been accepted or published.

- [M1] **U. Ikram** and I. Moon. "Joint Phase Reconstruction and Cell Segmentation from Raw Holograms for Per-Cell Dry-Mass Measurement." [Under review](#). [Code](#)
- [M2] **U. Ikram**, Y. Kim, and I. Moon. "Physics-Aware Lightweight Segmentation for Quantitative Holographic Cell Analysis with Optical Mass Preservation." Submitted to *Applied Physics B: Lasers and Optics*. [Under review](#). [Code](#)
- [M3] **U. Ikram**, Y. Kim, and I. Moon. "Delay-Aware Deployability of Multimodal Foundation Models for Real-Time Human-Robot Interaction." Submitted to *IEEE Transactions on Human-Machine Systems*. [Under review](#). [Code](#)
- [M4] **U. Ikram** (corresponding author), A. Jose, F. C. Eyzaguirre, and A. Mac Cawley. "Determinants of Glycaemic Outcomes in Children and Adolescents with Type 1 Diabetes: A Systematic Review and Evidence Map." Submitted to *Pediatric Diabetes*. PROSPERO CRD420251045872. [Under review](#).

RESEARCH EXPERIENCE

Intelligent Imaging and Vision Systems Laboratory, DGIST · Daegu, Republic of Korea Sep 2025 – present
Graduate Researcher, Department of Artificial Intelligence

Quantitative cell analysis in digital holographic microscopy

1. Physics-aware lightweight segmentation of quantitative phase images

Manuscript M2 | red blood cells, 1,090 annotated phase images | [code](#)

- Treated the predicted cell boundary as the domain over which phase is integrated into dry mass, and designed a Physics-Aware Phase Consistency Loss (phase-mask contrast, boundary-gradient alignment, integrated-phase preservation) on top of a class-weighted Dice loss. Used LoRA to adapt EdgeSAM, MobileSAM and a MobileNet-UNet from natural images to phase maps with four morphology classes.
- Against Dice-only training, integrated-phase error fell from 5.22% to 3.04% (EdgeSAM) and from 15.52% to 3.01% (MobileNet-UNet), with boundary F1 rising from 0.938 to 0.953 and from 0.913 to 0.970. The ablation showed the terms interact: phase-mask contrast alone collapsed boundary F1 to 0.386, and on MobileSAM the full objective was worse than a partial one.
- Compared rank-8 LoRA with full fine-tuning. Full fine-tuning drove echinocyte Dice to zero in all three backbones, while LoRA kept all four classes in both SAM-based backbones (EdgeSAM echinocyte Dice 0.776 against 0.000). The CNN had the best boundary F1 (0.971) while missing both minority classes, a direct case of an aggregate metric hiding class failure.

- Applied to 1,974 cells over an 11-timepoint, 47-day blood storage study: mean projected area fell from 63.1 to 43.4 μm^2 with relatively stable dry mass, and predicted dry mass agreed with annotation-derived values at $r = 0.966$ (median relative bias 1.5%). ONNX FP16 inference took 4.33 ms (228 FPS) at 16.4 MB peak memory on an RTX A5000; embedded deployment is not claimed.

2. Joint phase reconstruction and segmentation from raw holograms

Manuscript M1 | follow-on to M2, moving the phase reconstruction inside the learned model | [code](#)

- Built a 9.6M-parameter network (MobileNetV2 encoder, two U-Net decoders) that maps a raw digital hologram directly to quantitative phase and a cell segmentation, from which a fixed measurement chain computes per-cell area, circularity and dry mass with no numerical reconstruction at inference. Data: 800 fields of three cancer cell lines (control and four drugs), each with off-axis and in-line Gabor holograms.
- Reported paired per-cell dry-mass error with detection recall, a coverage-adjusted error and field totals, over three training seeds with a resolution criterion set in advance.
- Off-axis, matched-cell dry-mass error was 17.4% against 22.0% for the tested classical reconstruction pipeline at similar Dice (0.834 and 0.825); an exact domain-phase decomposition placed the error mainly in the phase. Detection recall (0.593; 0.912 for interior cells, 0.212 at the field boundary) is reported as the main limitation. In-line, one training run reached phase $r = 0.867$ where the tested classical in-line reconstruction recovered only near-background contrast.
- Tested measurement-aware integrated-phase losses and a hologram forward-model consistency term. Both integrated-phase terms increased per-cell dry-mass error beyond seed variation, and the forward-model term identified the propagation distance in-line but not off-axis; both are reported as negative results, with phase-derived labels stated as the main limitation.
- Showed without any trained model that a one-pixel boundary shift changes dry mass by 4.8% at Dice 0.974. Inference: 8.39 ms per 900×900 field (ONNX FP16, RTX A5000); a compact decoder with 65% fewer parameters was only 6–7% faster.

Multimodal foundation models under real-time constraints

3. Delay-aware deployability of vision-language models in a control loop

Manuscript M3 | CLIP, LoRA-CLIP and a Frozen Prefix-LM | [code](#)

- Treated model inference time as a transport delay in a wearable perception-in-the-loop system (PID-controlled mass-spring-damper plant) and derived a delay-dependent stability bound of 37.77 ms with a first-order Padé approximation and the Routh-Hurwitz criterion.
- Adapted CLIP (ViT-B/32) with LoRA (0.98M trainable parameters, 0.65% of the model) on a CC3M subset and evaluated zero-shot, few-shot ($k = 1$ to 16, three seeds) and linear-probe transfer on five datasets. Linear probing was nearly unchanged (85.69% against 85.95%) while zero-shot accuracy rose 6.05% on remote-sensing imagery and fell 4.37% on object-centric data, placing the change in the zero-shot readout rather than the representation. The generative Frozen Prefix-LM gave fluent output but reached only 51.25% under linear probing.
- Measured worst-case latency including jitter and a deployment margin: merged LoRA-CLIP reached 17.92 ms, inside both the nominal bound and the 18.76 ms bound under $\pm 50\%$ biomechanical variation, while unmerged adapters added 8.5 ms. The Frozen Prefix-LM took 442.02 ms, $11.7\times$ over the bound, because of autoregressive decoding rather than its 8.25 ms visual encoder.
- Software-in-the-loop simulation stayed bounded at 30 ms and diverged at 40 ms. Measured on a desktop RTX A5000; edge figures are projections, and no physical robot was used.

International collaboration · University of Limerick (Ireland) and Pontificia Universidad Católica de Chile

First and corresponding author

4. Evidence synthesis in paediatric type 1 diabetes

Manuscript M4 | PROSPERO-registered systematic review and evidence map

- Led a systematic review of 63 studies selected from 5,406 records across four databases, with two reviewers screening independently at every stage, appraisal with the Mixed Methods Appraisal Tool and reporting to PRISMA 2020.
- Built the PEARL framework: 297 factor-study associations organised into 47 determinants, 12 sub-domains and five domains under explicit assignment rules, with a counted weight of evidence for each. Found that consistency ran opposite to evidence volume: 68% of 148 physiological associations reached significance against 26 of 26 in the psychological domain, more plausibly explained by selective reporting than by larger effects.

PROFESSIONAL EXPERIENCE

- Developed end-to-end vision pipelines with YOLO-based object detection and region-of-interest segmentation for defence surveillance and situational awareness systems.
- Implemented and benchmarked autoencoder architectures for secure representation learning on multimodal data, so that encoded data stayed reliable and confidential in defence settings.
- Investigated neural compression and encryption methods that prevent adversarial decoding and unauthorised reconstruction of the data.
- Designed and integrated an LLM-based candidate evaluation system that uses prompt engineering to score and rank applicants, generate a technical quiz from each CV, and flag inconsistencies such as rapid job changes.
- Ran experiments on RK3588 SoC embedded boards, bringing on-device inference to low-power smart camera prototypes for edge applications.
- Worked with multidisciplinary teams to document research methods, contribute to design reviews, and propose optimisations for deploying AI on embedded hardware.

- Designed, built and trained deep learning models, including CNNs, LSTMs and autoencoders, for supply chain automation and industrial monitoring projects.
- Built neural classifiers on standardised accelerometer vibration data to tell healthy from faulty injection moulding machines, so maintenance could be planned before failures.
- Created an autoencoder that learns normal patterns in time series data and flags anomalies by their reconstruction error.
- Worked on machine positioning for supply chain tracking, measuring Bluetooth signal strength from IoT terminals on site and locating machines on the factory floor by trilateration.
- Analysed data transmission problems between field sensors and collection terminals, found the constraints, and improved the data flow so models were trained on the signal the sensors produced.
- Carried out statistical analysis, exploratory data analysis and feature engineering, and ran on-site experiments to collect labelled data for developing and testing algorithms.
- Prepared and presented business and technical reports to clients on quality assurance and performance improvements.

- Built responsive, accessible web interfaces in HTML and CSS, fixed front-end interface defects, and supported senior developers with site maintenance and performance monitoring.

UNDERGRADUATE RESEARCH PROJECTS

- **Explainable ensemble for knee osteoarthritis grading** | [code](#). Kellgren-Lawrence grading of 8,260 OAI knee radiographs with an ensemble of eight CNNs and mixed hard and soft voting (76.93% accuracy); Grad-CAM and LIME confirmed reliance on joint space narrowing.
- **Early detection of Alzheimer's disease (final-year thesis)** | [code](#). CNN classification, ridge regression for physiological scores and SVM models on ADNI data to identify patients likely to progress, deployed on a Jetson Nano.

TECHNICAL SKILLS

Programming	Python (primary), C++ , CUDA, MATLAB, SQL, TeX PyTorch (primary), TensorFlow, Hugging Face Transformers, PEFT, ONNX Runtime, OpenCV, scikit-learn
Holography and QPI	Off-axis and in-line hologram reconstruction (sideband filtering, angular-spectrum propagation, Gerchberg-Saxton phase retrieval), learned phase reconstruction, per-cell dry-mass measurement, forward-model consistency terms
Vision and multimodal	Semantic and instance segmentation, object detection, LoRA adaptation of CLIP and SAM-family models, zero-shot, few-shot and linear-probe evaluation, Grad-CAM and LIME
Evaluation	Per-class and per-cell metrics, multi-seed replication, bootstrap intervals, Bland-Altman agreement, error budgets, Padé and Routh-Hurwitz stability analysis, PRISMA 2020, PROSPERO, MMAT
Deployment	ONNX export and FP16 inference, latency and memory profiling, NVML energy measurement, embedded inference on RK3588 and Jetson Nano, Docker, Git
Languages	English (fluent), Urdu (native)